9. Combos and Blockers

We've done some intermittent mathematical work throughout the manual so far. I've taught the two most essential procedures of calculating required equity and required fold equity, but I've thus far stayed clear of what's by far the most useful mathematical tool in poker: understanding how to use combos and blockers to aid decision making.

We'll kick off by introducing the notion of the 'combo' and then move onto blockers later in the chapter.

A **Combo** is any combination of 2 of the 52 cards in the deck and constitutes an *exact* starting hand in Texas Holdem.

Ac Ah is one combo.
Ah As is another.
AA is the name of the group of all of the combos that contain two aces.

We say that **Ac Ah** is a combo of **AA**.

Before we learn how to apply combos to everyday poker thinking, it's important to memorise the three combo group types pre-flop and how many individual combos each one contains.

Group Type	Example Group	# Combos	Exact Combos
Pairs	88	6	8c8d 8c8h 8c8s 8d8h 8d8s 8h8s
Offsuit	AQo	12	AcQd AcQh AcQs AdQc AdQh AdQs AhQc AhQd AhQs AsQc AsQd AsQh
Suited	JTs	4	JcTc JdTd JhTh JsTs

Figure 47 - Hands and Combos

Note that **8c8d** is exactly the same combo as **8d8c**. We only care about which two cards make up the starting hand, not the order in which they are dealt. There are **1326** possible combos that Hero can be

dealt pre-flop, each one equally as likely as the next. However, when we start lumping combos into the groups we commonly refer to as 'starting hands' such as **KK**, **32o** or **45s**, we see that these hands are not dealt at equal frequencies. How abundant the combos of a hand are is the measure of how easy it is to be dealt that hand. For every pocket pair in a player's range pre-flop he will have 6 combos, but for every unpaired hand he'll have 16. It's more than twice as easy to be dealt an unpaired hand than it is a pair.

Say for example that some bizarre Villain in some bizarre poker universe had a BU opening range consisting of only **AA** and **72**, he would only have **AA** just 6 / (6 + 16) = **27%** of the time.

This goes to show that the natural distribution of starting hands is uneven. Appreciating the ways in which they are unevenly dealt gives us a better feel for how likely a player is to have certain parts of his range, or how balanced or unbalanced Hero's own range is when it contains certain starting hands. We can use combos for two purposes:

- To put our opponents on ranges and decide what we should do against those ranges.
- To build our own ranges so that we can aim at the appropriate long-term strategy for the situation.

We'll be using combos to build our own ranges in the next chapter on 3-betting. For now, let's take a look at some concrete examples of using combos to weight our opponents' ranges.

9.1 Using Combos Pre-Flop

The following hand illustrates a spot that we'll deal with more closely in Chapter 11 on reacting to 3-bets. For now we'll use this situation as a good example of how an appreciation of combos can help us avoid a very common pitfall that I often see among my students.

Hand 77

UTG – Unknown
HJ – Unknown
CO – Unknown
BU – Hero
SB – Passive Fish (44/12/**4**)
BB – Reg (24/20/**8**)

Dealt to Hero:

UTG folds, HJ folds, CO folds, **Hero raises to 3BB**, SB folds, **BB raises to 11BB**, **Hero raises to 24BB**, **BB raises to 100BB (all in)**, **Hero?**

Generally speaking, if Hero is going to 4-bet **QQ** BU vs BB it will be with the intention of calling a shove. Let's say for the sake of argument only that we know Villain's range is exactly **[KK+ AK]** when he shoves over the 4-bet. This could make 4-betting somewhat questionable in Hero's position if Villain doesn't flat many worse hands, but we'll ignore this for now.

Some students will now ask the following question:

"How can calling the shove with QQ be +EV? If Villain is only shoving* KK, AA *and* AK, *then surely Hero is either flipping or crushed?"

Okay before we go any further: what is the first thing Hero actually needs to consider in this situation? That's right, his required equity (RE) to make the call. If Hero fails to take this into account then he'll be on the wrong track from the start of his analysis. This is an End Of Action Spot (see Chapter 7)

We know that RE = ATC / (ATC + TP), which in this case is:

76 / (76 + 124.5) = **37.9%**.

The dangerous assumption to dismiss is that Hero needs to be good half of the time or more in order to call.

Now that we know our RE, we need to work out our equity against Villain's range. We'll do this first by using an equity calculator and then we'll come back to combos to explain why the calculation comes out as it does.

The calculator tells us that Hero's equity with **QQ** vs. the range of **[KK+ AK]** is **39.9%** and that this is indeed a comfortable call, but why is this the case?

It is true that vs. **AK** Hero has close to 50% equity and vs. **KK** and **AA** he has significantly less. My students' claim that Hero is 'flipping or crushed' is loosely correct. Hero's equity being sufficient to call here is a direct result of the combos in Villain's range weighting that range away from the bigger pairs that crush us and towards the **AK** portion against which we're a slight favourite.

Applying what we've learnt about combos so far, we know that there are 16 combos of unpaired hands (12 offsuit and 4 suited) and only 6 of a pocket pair. This means that Villain's range contains 16 combos of hands we are a slight favourite against (**AK**) and only twelve combos of hands that are crushing us (**KK** and **AA**). This is what gives us almost 40% equity here.

The flawed thought that here is that because there are two outcomes, one neutral and one bad, that the equity situation must be very bleak. The failure to realise the immensely different frequencies of occurrence of these two outcomes is something I call the **Equal Chance Fallacy.**

The **Equal Chance Fallacy** is to think that just because there are two options, those options are just as likely as each other.

When I leave my house to go for a run in a few minutes time we can stipulate two potential outcomes. It is certain that either of these two things will happen:

- **A:** I will be pecked to death by ravenous seagulls.
- **B:** I won't be pecked to death by ravenous seagulls.

I am almost entirely certain that B will occur and not A, and that's why I can go for a run without suicidal desires. The chances of A and B occurring are vastly different. There are far more ways that B can happen than A. In poker speak, there are far more combos of B than there are of A.

This might seem obvious and even insulting to your intelligence to state here, but I want to use the absurd seagull case to dispel what is actually a very common mistake made in poker thinking, namely, the disregard of combos and how likely they make one part of a range compared to another.

Let's move on to look at how we can use combos to decide how to play against our opponent's range post-flop.

9.2 Using Combos Post-Flop

Combos are also a natural weighting factor post-flop and determine how often our opponent will hold different parts of his range. One of the most essential skills in placing an opponent on an accurate range is being able to quickly estimate how likely one part of that range is compared to another. In Hand 78 below, Hero has the following two very useful player specific notes:

- Villain raised a flush draw on flop vs c-bet.
- Villain is capable of betting three streets as a bluff with a missed draw

Hand 78

UTG – Unknown
HJ – Unknown
CO – Unknown
BU – Hero
SB – Unknown
BB – Reg (20/17/**7**)

Dealt to Hero:

UTG folds, HJ folds, CO folds, **Hero raises to 2BB**, SB folds, **BB calls 1BB**.

Flop (4.5BB, 2 players)

BB checks, **Hero bets 2.5BB**, **BB raises to 8BB**, **Hero calls 5.5BB**.

Turn (20.5BB, 2 players)

BB bets 13.5BB, **Hero calls 13.5BB**.

River (47.5BB, 2 players)

BB bets 35BB, **Hero?**

First port of call as always in an end of action spot is to find RE which comes in at 35 / (35 + 82.5) =

29.7%

In game, we know that Villain's bet size was very close to 3/4 of the pot, which we know from our RE table (Figure 37) means that we need very close to 30% equity to call.

Let's try to put Villain on a likely range of hands:

When Villain raises the flop and blasts turn and river it's very likely that his range is polarised. It wouldn't make much sense to take a marginal value hand like **KJ** and overplay it to this extent as it's going to be tough to have good equity ***when called*** with such a hand.

When Hero has decent showdown value against a polarised betting range, he is always going to be winning against one part of Villain's range and losing against the other. We can therefore assign Villain two very different sub ranges relative to our hand.

- The very strong value hands that beat us.
- The bluffs that we beat.

Sub-Range 1: Value

What comprises this part of Villain's range? Hero can instantly discount **[QQ-AA]** and probably **JJ** too as it's just far too likely that a Reg with a 7% 3-bet stat will we 3-betting these hands pre-flop especially in late position battle spots such as BU vs. BB.

There is very little in the way of two pair in Villain's range. He would have to flat **J6s**, **J2s** or **62s** pre-flop and this is fairly unlikely from a tightish player even against the min raise. We'll exclude these hands from Villain's range.

So what does that leave for value? Essentially, just flopped sets. We've already discounted **JJ** due to pre-flop and the fact we hold a **J** in our hand makes this even less likely. That leaves just **66** and **22**. Villain can certainly have these hands. Let's count the combos.

Pre-flop these pocket pairs are 6 combos each, but what happens when we stick another of the same card on the flop? Well, it means that there are now only three suits of that card left to form combos. Since **6h** is now impossible for Villain to hold, he can only have **6c6d 6c6s 6d6s** for a total of 3 combos of **66**.

22 is even harder to have due to the second **2** appearing on the river. There is only 1 combo of that now (**2d2s**).

Villain's value range is only likely to contain just four value combos.

Sub-Range 2: Bluffs

The picture is already forming. If we only need 30% equity to call and Villain's value range is so tiny, it won't take many bluffs at all to be in there for calling to be +EV.

Let's be conservative and say that Villain only raises the flop with sets and flush draws; this assumes that he uses no bluff combos other than hands with two hearts in them. Let's be even more conservative and say that though we've seen Villain bluff missed flush draws before on the river he only bluffs with them half of the time.

Let's see how many combos of bluffs Villain can be estimated to have here. We'll start by seeing how many combos in his pre-flop calling range contain two hearts.

AA	AKs	AQs	AJs	ATs	A9s	A8s	A7s	A6s	A5s	A4s	A3s	A2s
AKo	KK	KQs	KJs	KTs	K9s	K8s	K7s	K6s	K5s	K4s	K3s	K2s
AQo	KQo	QQ	QJs	QTs	Q9s	Q8s	Q7s	Q6s	Q5s	Q4s	Q3s	Q2s
AJo	KJo	QJo	JJ	JTs	J9s	J8s	J7s	J6s	J5s	J4s	J3s	J2s
ATo	KTo	QTo	JTo	TT	T9s	T8s	T7s	T6s	T5s	T4s	T3s	T2s
A9o	K9o	Q9o	J9o	T9o	99	98s	97s	96s	95s	94s	93s	92s
A8o	K8o	Q8o	J8o	T8o	98o	88	87s	86s	85s	84s	83s	82s
A7o	K7o	Q7o	J7o	T7o	97o	87o	77	76s	75s	74s	73s	72s
A6o	K6o	Q6o	J6o	T6o	96o	86o	76o	66	65s	64s	63s	62s
A5o	K5o	Q5o	J5o	T5o	95o	85o	75o	65o	55	54s	53s	52s
A4o	K4o	Q4o	J4o	T4o	94o	84o	74o	64o	54o	44	43s	42s
A3o	K3o	Q3o	J3o	T3o	93o	83o	73o	63o	53o	43o	33	32s
A2o	K2o	Q2o	J2o	T2o	92o	82o	72o	62o	52o	42o	32o	22

Bluff Range Value Range

Figure 48 - Ranging: Comparing Value to Bluffs

In green are the flopped sets (now boats or Quads) that we already decided were very likely to be a

part of Villain's range.

In red is an estimate of all the suited hands that Villain flats pre-flop. Note that the very best and very worst suited hands have been excluded on the grounds that they are too good and too bad to flat the open with respectively.

I've also removed any hands containing **Jh** as this would not be part of a polarised range on this board given the SDV of the hand. I've also removed any hands containing a **6** or a **2** as it's not possible to flop a flush draw with these given the **6h** and **2h** are on the flop. Now there are four combos of each suited hand in this range, but only one of each (**hearts**) is going to give Villain a flush draw.

We said we'd use half of all the possible flopped flush draws for Villain's river bluffing range, which is fairly conservative given our two notes on this player.

In total, there are 31 combos of flush draws in Villain's range. We'll therefore estimate that Villain has 15 bluff combos on the river.

The results are staggering. If our estimates are not miles off, then Villain is only betting a better hand then Hero's 4 / 19 = 21% of the time. Since Hero beats all of Villain's bluffs he beats 15 /19 combos and can expect to have **79%** equity vs this range.

Hero only needs 30% equity to call and so after this analysis we see that Hero's decision isn't close.

Hero calls 35BB.

Balance Considerations

A **balanced river betting range** should grant Villain's bluff catchers exactly his RE% in order to make calling break even for Villain. This ensures there is no exploitative strategy available to him.

Remember in Chapter 8 when we talked about whether to have a raising range or not on a certain flop? If Villain wants a balanced strategy in Hand 78 above, then it will be very difficult to correctly balance a river betting range between value and bluffs. He has just four value combos and so would need to bluff even less combos of his missed draws to be unexploitable to an overcalling strategy from Hero.

To be balanced, Villain would need to make Hero ***indifferent*** to calling with a bluff catcher, which is to make it 0EV for Hero to call with a hand like the **AJ** he holds in Hand 78. This would require giving Hero exactly **30%** equity and no more or less as this is Hero's RE due to Villain's sizing. To do

this Villain would need to bluff 30% of the time and value bet 70% of the time. Since he only has four value combos, even having two bluff combos would equate to Hero having 2/6 = **33%** equity and the option of profitably overcalling his range on the river.

Combos naturally skew ranges. It's much easier to get to the river with a busted flush draw than with a full house or better. In Hand 77 Villain is getting exploited enormously by Hero's plan to overcall his range on this river. Hero should be calling any hand with more than 30% equity against Villain's range and that will be almost his entire range for facing this bet. Villain is getting strategically crushed thanks to Hero's reads and according exploitative plan.

Many Villains, even Regs, will be obviously unbalanced in this way for no good reason and this is exactly why we can beat them. We'll be returning to this theme in Chapter 12 when we go on to talk about forming our own post-flop bluffing ranges. This will save us from suffering the same fate as Villain in that last hand.

9.3 Blockers

In Hand 78 it became harder for Villain to hold either **66** or **22** when the flop came down with both a **6** and a **2**. We can say that these hands were 'blocked' by the flop; the combos of each were slashed from 6 to 3. If we'd thought that Villain could have had **JJ** even though he didn't 3-bet pre-flop, we would have discounted this to only 1 combo because the hand was not only blocked by the flop, but by Hero's **AJ** too.

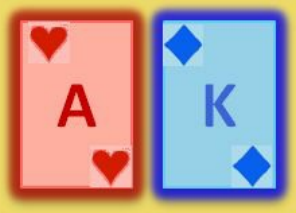
A **Blocker** is a card in either a player's hand or on the flop that reduces the number of combinations of some hands in another player's range.

Holding an **A** reduces the combinations of:
AK from 16 to 12
AKo from 12 to 9
AKs from 4 to 3
AA from 6 to 3
A set of AA on Axx from 3 to 1

The following hand is a pre-flop spot in which blockers are crucial to our equity being as good as it is.

Hand 79

UTG – Unknown
HJ – Unknown
CO – Unknown
BU – Hero
SB – Passive Fish (44/12/**4**)
BB – Reg (24/20/**8**)

Dealt to Hero: 

UTG folds, HJ folds, CO folds, **Hero raises to 3BB**, SB folds, **BB raises to 11BB**, **Hero raises to 24BB**, **BB raises to 100BB (all in)**, **Hero?**

AKo is a very standard 4-bet call BU vs. BB against the average active Reg. The reason for this is two fold.

1. Fold Equity

Hero picks up a lot of dead money when Villain folds to the 4-bet. Villain folds to the 4-bet more than usual when Hero holds both an **A** and a **K** as these blockers eliminate a lot of **AA**, **KK** and **AK** combos which obviously don't fold to the 4-bet.

Let's assume that Villain 3-bet shoves a range of **[JJ+, AK, AQs]** which is a reasonable, but tightish range to shove over a BU 4-bet Reg vs. Reg. If Hero's cards were **4c4d** then this shoving range would contain **44 combos** (24 pairs + 16 **AK** + 4 **AQs**).

As it is, when Hero has **AKo**, he blocks three **AA** combos, three **KK** combos and seven **AK** combos. Villain's 3-bet shoving range has just 44 - 13 = **31 combos** now instead of 44. Hero's blockers ensure that Villain only shoves 31/(X+31) % of time instead of 44/(X+44) % of the time, where X is Villain's amount of 3-bet bluff combos. The first number will always be lower because regardless of how many bluffs Villain uses, his value hands are blocked to a greater extent. This added fold equity helps Hero's EV tremendously.

2. Equity

The **A** and **K** blockers don't just boost the EV of 4-bet/calling by increasing Hero's FE, they also help his pot equity. The only hands in Villain's range that really have **AKo** in bad shape are **KK** and **AA**. Hero's **AKo** is a self protecting sledgehammer in that it blocks the hands that hurt it while grabbing a great deal of fold equity and pot equity vs. the rest of Villain's range.

Hero's equity vs. Villain's estimated shoving range of **[JJ+ AK AQs]** is **42%** and this is plenty for Hero to call off the rest of his chips after 4-betting. RE here is **37.9%**. This equity would be far less if Hero's blockers didn't stop Villain from having four aces and kings to build nutted hands with.

You might be wondering why Hero wants to get all in in this situation in the first place if he only has 42% equity vs. Villain's getting it in range. It must be -EV surely?

No. The fold equity that Hero gains by 4-betting and making Villain fold the rest of his range more than compensates for the deficit in overall pot equity when the money goes in. Moreover, Villain's range can and should be wider than this in many situations, which will see Hero's pot equity climb further. Finally, it's important for Hero to have **AK** in his 4-bet/call range for balance reasons. Undoubtedly, Hero will want to be 4-bet bluffing and folding to a 5-bet with a sizable range here and so will need those precious 16 combos of **AK** that can 4-bet/call to prevent Villain from having very profitable light 5-bet shoves at his disposal.

We'll deal with building 4-bet ranges in Chapter 11 when we come to face 3-bets. For now, it's time to be on the other end of the battle and take a plunge into how to assess life as the 3-bettor. Make sure you're very comfortable with the notions of combos and blockers discussed in this chapter before we jump right in.

There are post-flop uses of blockers too and these will be discussed later in the manual when we come to using blockers to create balanced bluffing ranges and calling down ranges. In the meantime, feel free to have a think about how they might be used in these ways.